BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE APPLICATIONS AND GEO-INFORMATICS

WEEKLY PROGRESS REPORT (8/06/2026 – 14/06/2026)

WEEK 4

PROJECT NAME	ROAD CONDITION DETECTION

PROJECT DESCRIPTION :	**AUTOMATIC DETECTION AND SEGMENTATION OF ROAD DAMAGES**
GROUP MEMBER :	**Omana Arya, Anshika Tiwari**
GROUP ID :	**2026G263**
GROUP GUIDE :	**MR. MANOJ PANDYA**
GROUP COORDINATOR :	**MR. MANOJ PANDYA**
COLLEGE GUIDE NAME :	**DR. SHWETA SAHARAN**
COLLEGE GUIDE No. :	**95096 76076**
COLLEGE GUIDE EMAIL. :	**shweta.saharan@gsv.ac.in**
COLLEGE NAME :	**GATI SHAKTI VISHWAVIDYALAYA, VADODARA**

Date	Task
8/06/2026	Collected additional road-condition video data, extracted frames from village-road videos, and prepared data for dataset enhancement.
9/06/2026	Updated and organized the road damage segmentation dataset, verified annotations, and incorporated new samples to improve class representation.
10/06/2026	Trained the YOLO11l segmentation model on the updated dataset, monitored training progress, and analyzed validation metrics.
11/06/2026	Evaluated model performance on unseen road videos, investigated missed detections for cracks and ravelling, and studied methods to improve generalization.
12/06/2026	Explored inference pipeline improvements, checkpoint management, video preprocessing techniques, and planned dashboard development for visualizing road-condition analysis results.
13/06/2026	**Holiday**
14/06/2026	**Holiday (Sunday)**

Next Week Plan	Develop the Streamlit dashboard, integrate model inference and road-condition statistics, perform final testing on unseen videos, and prepare the project report and presentation.

REFERENCE:

1. **Ultralytics YOLO11. *YOLO11 Documentation.* Available at: https://docs.ultralytics.com/**
2. **PyTorch. *PyTorch Documentation.* Available at: https://pytorch.org/docs/stable/**
3. **OpenCV. *OpenCV Documentation.* Available at: https://docs.opencv.org/**

4. **Streamlit.** ***Streamlit Documentation.*** **Available at:** https://docs.streamlit.io/

5. **Roboflow.** ***Roboflow Documentation.*** **Available at:** https://docs.roboflow.com/

6. **NumPy.** ***NumPy Documentation.*** **Available at:** https://numpy.org/doc/

7. **Pandas.** ***Pandas Documentation.*** **Available at:** https://pandas.pydata.org/docs/

8. **Matplotlib.** ***Matplotlib Documentation.*** **Available at:** https://matplotlib.org/stable/

SCREEN SHOTS: